

Celebrity generation experiment

The images shown in Fig. 3 are from a variational autoencoder that was trained on the CelebA dataset²³. This dataset consists of 202,599 face images of dimension $178 \times 218 \times 3$, of which 162,770 images were used as the training set, and 19,962 images were used as the test set (the remaining 19,867 validation images were not used). The images were scaled down to $64 \times 64 \times 3$ via `torch.resize()` to speed up training and make experimentation more feasible.

The VAE uses 5 convolutional layers for the encoder (3, 32, 64, 128, 256, 512) with kernel size 4, stride 2, padding 1, and the same layers in reverse as transposed convolutions for the decoder. A latent dimension of 512 was used. Training used binary cross-entropy loss paired with the ADAM optimizer. Training was carried out for 1000 epochs, using a learning rate of $1e-4$ and a batch size of 64.

FPGA generation experiment

A Variational Autoencoder (VAE) is trained to generate the MNIST digits following the architecture in Fig. 9c. The deterministic and probabilistic versions of this design are implemented on a ZCU104 board, which houses a Zynq Ultrascale+ ZU7EV FPGA. The weights for all models are converted to fixed-point Q6.25. The p-bit design uses a single sample, which proved sufficient for generating the images. The design was run at 100 MHz. The input is deterministically binarized before feeding into the first layer of the model. The full VAE was trained using a GPU, and the decoder was implemented on the FPGA. For the MAC operation, we implement a high precision intermediate accumulation as is standard in FMAC units.

Data availability

Code for these experiments is available upon request from the corresponding author. All data used were part of publicly available ML datasets.

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Author contributions

L.G. set up the experiments and machine learning pipelines, conducted and recorded training and inference experiments, and built the FPGA design. M.L. and S.S. provided an energy breakdown of a QMC-ASIC and conducted additional simulations to get energy estimates for core operations on a TSMC-65nm platform. All authors were critical in establishing the probabilistic methods discussed and demonstrated in the manuscript. L.G. and S.D. lead the writing of the paper, with contributions/revisions from all authors. All authors read and approved the final manuscript.

Competing interests

L.G., S.D., and B.B. have financial interests in Ludwig Computing. S.S. has a financial interest in Ixana.

Additional information

Supplementary information The online version contains supplementary material available at

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